Stelios Zacharioudakis

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SUMMARY

Machine learning engineer with hands-on ownership of a production association platform. I build and deploy AI agents, multilingual LLM workflows and ML systems, with experience in reinforcement learning, GPU inference benchmarking and live data pipelines.

PROFESSIONAL EXPERIENCE

Paphos Medical Association **Jun 2022 - Jul 2026**
Head Engineer (Pro Bono)
- Designed, built and donated AsklepiosMed, a production platform for member services and association administration.
- Built React interfaces, Express APIs and PostgreSQL workflows with role-based access for members and staff. Owned deployment on a Linux VPS.

SELECTED ENGINEERING PROJECTS

AsklepiosMed - Multilingual NLP workflows Case study
- Extended AsklepiosMed after my pro bono role ended, integrating LLM-assisted drafting, email preparation and proofreading into the administrative publishing workflow.
- Built translation workflows from Greek into English, Russian and Simplified Chinese, with terminology checks, output validation, fallback handling and editorial review.

ForgeRL / ForgeBench - Reinforcement learning and agent evaluation Agent evaluation demo
- Built a sandboxed coding-agent benchmark with an offline fitted-Q routing controller, persistent budget controls and replayable traces. Compared six policies in a 162-episode transfer pilot and ran a separate 1,296-attempt follow-up.

Inference Systems Lab - GPU serving evaluation Code and results
- Benchmarked 16,384 vLLM requests across RTX 4090 and A100 GPUs, two model sizes and four repeats. Shared-prefix caching yielded 1.23–1.47× throughput at concurrency 32 in this benchmark. Tracked output mismatches alongside performance.

MTA-Scan - Streaming data and temporal evaluation Live dashboard
- Built a monitored GTFS-Realtime pipeline and FastAPI/Mapbox workspace with feed freshness checks, forecast expiry and an archive of official alert revisions. Verified 44 daily exports and evaluated arrival-spacing proxy forecasts across 28 days.

MRI reconstruction - PyTorch and generative models Thesis and evidence
- BSc thesis: Excellent, 10/10
- Reimplemented a published score-based reconstruction method with Flax-to-PyTorch checkpoint conversion; reproduced reference PSNR within 0.15 dB on 256 BraTS slices at 4x, 8x and 24x undersampling.
- Evaluated reconstruction quality, uncertainty and retrospective domain shift; published reproducible evidence and independent CPU ensemble diagnostics.

TECHNICAL SKILLS

ML & AI: Python, PyTorch, reinforcement learning, LLM agents, world models, generative models, NLP, vLLM, NumPy, SciPy, scikit-learn, experimental evaluation
Engineering: FastAPI, React, Node.js/Express, JavaScript, PostgreSQL, Redis, Linux, Nginx, Docker, Git, automated testing

EDUCATION

BSc, Computer Science **Jun 2026**
National and Kapodistrian University of Athens
- Degree: Very Good | Thesis: Excellent (10/10), deposited September 2026.
- Independent technical work: verifier-guided self-improvement, continual learning for coding agents, latent world models and policy evaluation; ongoing and unpublished.